

Homework Problems
Physics 451/551
Due: September 11, 2018

Solve and submit 3 of these problems from Chapter 4

1. Fetter and Walecka, Problem 4.1
2. Fetter and Walecka, Problem 4.3
3. Fetter and Walecka, Problem 4.4
4. Fetter and Walecka, Problem 4.6
5. Fetter and Walecka, Problem 4.8

4.1 Let θ_1 be the angle the hoop makes with the vertical, and θ_2 the angle the particle makes with the (moving) vertical from the center of the hoop.

- a. The moment of inertia for the hoop around its center is MR^2 . By the parallel axis theorem the moment of inertia about the edge is $MR^2 + MR^2 = 2MR^2$ and the kinetic energy of the hoop is

$$T_1 = \frac{1}{2} I \dot{\theta}_1^2 = \frac{2MR^2 \dot{\theta}_1^2}{2}.$$

The position of the center of mass (origin at the fixed point in the hoop) is

$$\vec{x}_{cm} = R \sin \theta_1 \hat{x} - R \cos \theta_1 \hat{y}.$$

The position of the particle is $\vec{x} = \vec{x}_{cm}(t) + R \sin \theta_2 \hat{x} - R \cos \theta_2 \hat{y}$. Differentiation of this equation by time and taking the small angle approximation yields

$$T_2 = \frac{M}{2} R^2 (\dot{\theta}_1 + \dot{\theta}_2)^2$$

The gravitational potential of the hoop is

$$U_1 = Mgy_{cm} = -MgR \cos \theta_1 \approx C + \frac{MgR}{2} \theta_1^2$$

where C is an irrelevant constant. Remembering that the displacement of the particle is from both the displacement of the center of mass of the hoop and from the angle with the vertical,

$$U_2 = -MgR \cos \theta_1 - MgR \cos \theta_2 \approx \frac{MgR}{2} [\theta_1^2 + \theta_2^2],$$

again ignoring an irrelevant constant.

The Euler-Lagrange equations of motion are

$$\frac{d}{dt} \begin{bmatrix} 3MR^2 & MR^2 \\ MR^2 & MR^2 \end{bmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{pmatrix} + \begin{pmatrix} 2MgR & 0 \\ 0 & MgR \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = 0.$$

The equation for the normal mode frequencies is

$$\det \begin{pmatrix} 2MgR - 3MR^2 \omega^2 & -MR^2 \omega^2 \\ -MR^2 \omega^2 & MgR - MR^2 \omega^2 \end{pmatrix} = 0.$$

The "quadratic" (in the square of the angular frequency) formula to solve is

$$2M^2 R^4 \omega^4 - 5MR^2 MgR \omega^2 + 2M^2 g^2 R^2 = 0.$$

The solution for the angular frequency squared is

$$\omega^2 = \frac{5MR^2 MgR \pm \sqrt{5^2 - 16MR^2 MgR}}{4M^2 R^4} = \frac{2g}{R}, \frac{g}{2R},$$

as given in the problem.

- b) The first eigenvector solves

$$\begin{pmatrix} -4 & -2 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = 0 \rightarrow v_+ = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

It is anti-symmetric motion with the particle moving twice the angle of the hoop. The

second eigenvector solves

$$\begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = 0 \rightarrow v_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

It is a symmetrical motion where the particle does not move with respect to the hoop.
Normalizing with the mass matrix

$$MR^2 \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow v_+ = \frac{1}{\sqrt{3MR^2}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}, v_- = \frac{1}{\sqrt{6MR^2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Note that the eigenvectors are indeed normal to each other when multiplied through the mass matrix.

c) The modal matrix is

$$\frac{1}{\sqrt{MR^2}} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} \\ -2/\sqrt{3} & 1/\sqrt{6} \end{pmatrix}$$

d) Diagonalizing the mass matrix gives the unit matrix back

$$\begin{aligned} & \frac{1}{\sqrt{MR^2}} \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} MR^2 \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{MR^2}} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} \\ -2/\sqrt{3} & 1/\sqrt{6} \end{pmatrix} = \\ & = \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 4/\sqrt{6} \\ -1/\sqrt{3} & 2/\sqrt{6} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Diagonalizing the v matrix gives the eigenvalues back

$$\begin{aligned} & \frac{1}{\sqrt{MR^2}} \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} MgR \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{\sqrt{MR^2}} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} \\ -2/\sqrt{3} & 1/\sqrt{6} \end{pmatrix} = \\ & = \frac{g}{R} \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} 2/\sqrt{3} & 2/\sqrt{6} \\ -2/\sqrt{3} & 1/\sqrt{6} \end{pmatrix} = \frac{g}{R} \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} \end{aligned}$$

Finally, it should be noted that the problem could have just as well be solved using the variable $\theta'_2 = \theta_2 - \theta_1$, which is the angle of the particle with respect to the rotating hoop. The eigenvalues will be the same. The eigenvectors will be adjusted. In particular, the plus eigenvector will have no θ'_2 component!

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Small Oscillations

1. Fetter & Walecka, Problem 4.3.

a) Introduce the small transverse displacements around the equilibrium $\eta_1 \approx \ell\theta_1$ and $\eta_2 \approx \ell\theta_2$ in the full Lagrangian:

$$\begin{aligned}
 L &= T_1 + T_2 - V_1 - V_2 \\
 &= \frac{1}{2}m_1\ell^2\dot{\theta}_1^2 + \frac{1}{2}m_2(\ell\dot{\theta}_1 + \ell\dot{\theta}_2)^2 - m_1g\ell(1 - \cos\theta_1) - m_2g\ell(1 - \cos\theta_1) - m_2g\ell(1 - \cos\theta_2) \\
 &= \frac{1}{2}m_1\ell^2\dot{\theta}_1^2 + \frac{1}{2}m_2(\ell\dot{\theta}_1 + \ell\dot{\theta}_2)^2 - \frac{1}{2}m_1g\ell\theta_1^2 - \frac{1}{2}m_2g\ell\theta_1^2 - \frac{1}{2}m_2g\ell\theta_2^2 \\
 &= \frac{1}{2}m_1\dot{\eta}_1^2 + \frac{1}{2}m_2(\dot{\eta}_1 + \dot{\eta}_2)^2 - \frac{g}{2\ell}m_1\eta_1^2 - \frac{g}{2\ell}m_2\eta_1^2 - \frac{g}{2\ell}m_2\eta_2^2 \\
 &= \frac{1}{2}m_1\dot{\eta}_1^2 + \frac{1}{2}m_2(\dot{\eta}_1 + \dot{\eta}_2)^2 - \frac{g}{2\ell}[(m_1 + m_2)\eta_1^2 + m_2\eta_2^2].
 \end{aligned}$$

b) The mass and potential matrices are

$$\begin{aligned}
 M &= \begin{bmatrix} m_1 + m_2 & m_2 \\ m_2 & m_2 \end{bmatrix}, \\
 V &= \frac{g}{\ell} \begin{bmatrix} m_1 + m_2 & 0 \\ 0 & m_2 \end{bmatrix}.
 \end{aligned}$$

The eigenvalue problem then becomes

$$\begin{vmatrix} \frac{g}{\ell}(m_1 + m_2) - \omega^2(m_1 + m_2) & -\omega^2 m_2 \\ -\omega^2 m_2 & \frac{g}{\ell}m_2 - \omega^2 m_2 \end{vmatrix} = 0.$$

This becomes now the quadratic equation in ω^2

$$(m_2(m_1 + m_2) - m_2^2)\omega^4 - 2\frac{g}{\ell}m_2(m_1 + m_2)\omega^2 + \frac{g^2}{\ell^2}m_2(m_1 + m_2) = 0$$

or

$$\frac{m_1}{m_1 + m_2}\omega^4 - 2\frac{g}{\ell}\omega^2 + \frac{g^2}{\ell^2} = 0.$$

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With $\gamma^2 = \frac{m_2}{m_1+m_2}$ we find the positive discriminant

$$D = 4 \left(1 - \frac{m_1}{m_1+m_2} \right) \frac{g^2}{\ell^2} = 4 \frac{m_2}{m_1+m_2} \frac{g^2}{\ell^2} = 4\gamma^2 \frac{g^2}{\ell^2}$$

and the eigenvalues are

$$\begin{aligned} \omega_{\pm}^2 &= \frac{1}{2} \frac{m_1+m_2}{m_1} \left(2\frac{g}{\ell} \pm 2\gamma\frac{g}{\ell} \right) \\ &= \frac{g}{\ell} \frac{1 \pm \gamma}{1 - \gamma^2} \\ &= \frac{g}{\ell} \frac{1}{1 \mp \gamma} \end{aligned}$$

where we used $\frac{m_1}{m_1+m_2} = 1 - \gamma^2$.

c) To determine the eigenvectors we try to find $z_{\pm}$ from $(V - \omega_{\pm}^2 M)z_{\pm} = 0$, or

$$\begin{aligned} \begin{bmatrix} (m_1+m_2) \left(\frac{g}{\ell} - \omega_{\pm}^2 \right) & -m_2 \omega_{\pm}^2 \\ -m_2 \omega_{\pm}^2 & m_2 \left(\frac{g}{\ell} - \omega_{\pm}^2 \right) \end{bmatrix} z_{\pm} &= 0 \\ \begin{bmatrix} (m_1+m_2) \frac{g}{\ell} \left(1 - \frac{1}{1 \mp \gamma} \right) & -m_2 \frac{g}{\ell} \frac{1}{1 \mp \gamma} \\ -m_2 \frac{g}{\ell} \frac{1}{1 \mp \gamma} & m_2 \frac{g}{\ell} \left(1 - \frac{1}{1 \mp \gamma} \right) \end{bmatrix} z_{\pm} &= 0 \\ m_2 \frac{g}{\ell} \frac{1}{1 \mp \gamma} \begin{bmatrix} \frac{m_1+m_2}{m_2} (1 \mp \gamma - 1) & -1 \\ -1 & 1 \mp \gamma - 1 \end{bmatrix} z_{\pm} &= 0 \\ -m_2 \frac{g}{\ell} \frac{1}{1 \mp \gamma} \begin{bmatrix} \pm \frac{1}{\gamma} & 1 \\ 1 & \pm \gamma \end{bmatrix} z_{\pm} &= 0. \end{aligned}$$

The eigenvectors are then

$$z_{\pm} = \begin{bmatrix} \mp \gamma \\ 1 \end{bmatrix},$$

or, with

$$\begin{aligned} z_{\pm}^T M z_{\pm} &= \begin{bmatrix} \mp \gamma & 1 \end{bmatrix} \begin{bmatrix} m_1+m_2 & m_2 \\ m_2 & m_2 \end{bmatrix} \begin{bmatrix} \mp \gamma \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \mp \gamma & 1 \end{bmatrix} \begin{bmatrix} \mp \gamma(m_1+m_2) + m_2 \\ \mp \gamma m_2 + m_2 \end{bmatrix} \\ &= \gamma^2(m_1+m_2) \mp \gamma m_2 \mp \gamma m_2 + m_2 \\ &= 2m_2(1 \mp \gamma), \end{aligned}$$

we find the normalized eigenvectors

$$\tilde{z}_{\pm} = (2m_2(1 \mp \gamma))^{-1/2} \begin{bmatrix} \mp \gamma \\ 1 \end{bmatrix},$$

- For a very large middle mass, $m_1 \gg m_2$, we have $\gamma = \sqrt{m_2/m_1}$ small. Both frequencies are $\omega_{\pm} \approx \frac{g}{\ell}$ and the eigenvectors are

$$z_{\pm} = \begin{bmatrix} \pm m_2/m_1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Only the lowest mass is moving with the frequency of a pendulum of length ℓ .

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- For a very small middle mass, $m_1 \ll m_2$, we have $\gamma^2 \approx 1$. The eigenvectors are then

$$z_{\pm} = \begin{bmatrix} \pm 1 \\ 1 \end{bmatrix}.$$

The frequency ω_- is large and the eigenvector is

$$z_- = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

This corresponds to the middle mass vibrating quickly under the large tension in the string. The other frequency is $\omega_+ = \frac{g}{2\ell}$ and the corresponding eigenvector is

$$z_+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

This describes the motion of a pendulum that is twice as long and with both masses moving in the same direction.

d) The modal matrix is

$$\begin{aligned} U &= [\tilde{z}_- \quad \tilde{z}_+] \\ &= (2m_2)^{-1/2} \begin{bmatrix} \frac{\gamma}{\sqrt{1+\gamma}} & -\frac{\gamma}{\sqrt{1-\gamma}} \\ \frac{1}{\sqrt{1+\gamma}} & \frac{1}{\sqrt{1-\gamma}} \end{bmatrix}. \end{aligned}$$

To simplify we use

$$\begin{aligned} \frac{1}{\sqrt{1 \pm \gamma}} &= \sqrt{1 \mp \gamma} \frac{1}{\sqrt{1 - \gamma^2}} \\ &= \sqrt{1 \mp \gamma} \sqrt{\frac{m_1 + m_2}{m_1}} \\ &= \sqrt{1 \mp \gamma} \sqrt{\frac{m_2}{m_1}} \frac{1}{\gamma} \end{aligned}$$

to find

$$U = (2m_1)^{-1/2} \begin{bmatrix} \sqrt{1-\gamma} & -\sqrt{1+\gamma} \\ \gamma^{-1}\sqrt{1-\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix}$$

This does indeed diagonalize M , by construction, and V :

$$\begin{aligned} U^T V U &= \frac{1}{2m_1} \frac{g}{\ell} \begin{bmatrix} \sqrt{1-\gamma} & \gamma^{-1}\sqrt{1-\gamma} \\ -\sqrt{1+\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \begin{bmatrix} m_1 + m_2 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \sqrt{1-\gamma} & -\sqrt{1+\gamma} \\ \gamma^{-1}\sqrt{1-\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \\ &= \frac{1}{2m_1} \frac{g}{\ell} \begin{bmatrix} \sqrt{1-\gamma} & \gamma^{-1}\sqrt{1-\gamma} \\ -\sqrt{1+\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \begin{bmatrix} (m_1 + m_2)\sqrt{1-\gamma} & -(m_1 + m_2)\sqrt{1+\gamma} \\ m_2\gamma^{-1}\sqrt{1-\gamma} & m_2\gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \\ &= \frac{1}{2m_1} \frac{g}{\ell} \begin{bmatrix} (m_1 + m_2)(1-\gamma) + m_2\gamma^{-2}(1-\gamma) & -(m_1 + m_2)\sqrt{1-\gamma^2} + m_2\gamma^{-2}\sqrt{1-\gamma^2} \\ -(m_1 + m_2)\sqrt{1-\gamma^2} + m_2\gamma^{-2}\sqrt{1-\gamma^2} & (m_1 + m_2)(1+\gamma) + m_2\gamma^{-2}(1+\gamma) \end{bmatrix} \\ &= \frac{1}{2m_1} \frac{g}{\ell} \begin{bmatrix} 2(m_1 + m_2)(1-\gamma) & 0 \\ 0 & 2(m_1 + m_2)(1+\gamma) \end{bmatrix} \\ &= \frac{g}{\ell} \begin{bmatrix} \frac{1}{1+\gamma} & 0 \\ 0 & \frac{1}{1-\gamma} \end{bmatrix} = \begin{bmatrix} \omega_-^2 & 0 \\ 0 & \omega_+^2 \end{bmatrix}. \end{aligned}$$

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e) The normal coordinates $\xi_{\pm}$ are

$$\begin{aligned}
 \begin{bmatrix} \xi_- \\ \xi_+ \end{bmatrix} &= U^T M \eta \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma} & \gamma^{-1}\sqrt{1-\gamma} \\ -\sqrt{1+\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \begin{bmatrix} m_1+m_2 & m_2 \\ m_2 & m_2 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma} & \gamma^{-1}\sqrt{1-\gamma} \\ -\sqrt{1+\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \begin{bmatrix} (m_1+m_2)\eta_1 + m_2\eta_2 \\ m_2\eta_1 + m_2\eta_2 \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma}((m_1+m_2)\eta_1 + m_2\eta_2 + \gamma^{-1}(m_2\eta_1 + m_2\eta_2)) \\ \sqrt{1+\gamma}(-(m_1+m_2)\eta_1 - m_2\eta_2 + \gamma^{-1}(m_2\eta_1 + m_2\eta_2)) \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma}((m_1+m_2)\eta_1 + \gamma^2(m_1+m_2)\eta_2 + \gamma(m_1+m_2)(\eta_1 + \eta_2)) \\ \sqrt{1+\gamma}(-(m_1+m_2)\eta_1 - \gamma^2(m_1+m_2)\eta_2 + \gamma(m_1+m_2)(\eta_1 + \eta_2)) \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma}(m_1+m_2)(\eta_1 + \gamma^2\eta_2 + \gamma(\eta_1 + \eta_2)) \\ \sqrt{1+\gamma}(m_1+m_2)(-\eta_1 - \gamma^2\eta_2 + \gamma(\eta_1 + \eta_2)) \end{bmatrix} \\
 &= \frac{m_1+m_2}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma}(1+\gamma)(\eta_1 + \gamma\eta_2) \\ -\sqrt{1+\gamma}(1-\gamma)(\eta_1 - \gamma\eta_2) \end{bmatrix} \\
 &= \frac{\sqrt{m_1}}{\sqrt{2}(1-\gamma^2)} \begin{bmatrix} \sqrt{1-\gamma}(1+\gamma)(\eta_1 + \gamma\eta_2) \\ -\sqrt{1+\gamma}(1-\gamma)(\eta_1 - \gamma\eta_2) \end{bmatrix} \\
 &= \frac{\sqrt{m_1}}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{1-\gamma}}(\eta_1 + \gamma\eta_2) \\ -\frac{1}{\sqrt{1+\gamma}}(\eta_1 - \gamma\eta_2) \end{bmatrix}
 \end{aligned}$$

Inversely, the coordinates η in terms of the normal coordinates are

$$\begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = U \xi = \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \sqrt{1-\gamma} & -\sqrt{1+\gamma} \\ \gamma^{-1}\sqrt{1-\gamma} & \gamma^{-1}\sqrt{1+\gamma} \end{bmatrix} \begin{bmatrix} \xi_- \\ \xi_+ \end{bmatrix}$$

f) The initial conditions are $\eta_1(0) = a$ and $\eta_2(0) = 0$, and $\dot{\eta}_1 = \dot{\eta}_2 = 0$. The general solution

$$\begin{bmatrix} \xi_- \\ \xi_+ \end{bmatrix} = \begin{bmatrix} C_- \cos(\omega_- t + \phi_-) \\ C_+ \cos(\omega_+ t + \phi_+) \end{bmatrix}$$

gives the initial conditions on the normal coordinates,

$$\begin{bmatrix} \xi_-(0) \\ \xi_+(0) \end{bmatrix} = \begin{bmatrix} C_- \cos \phi_- \\ C_+ \cos \phi_+ \end{bmatrix} = \frac{\sqrt{m_1}}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{1-\gamma}}a \\ -\frac{1}{\sqrt{1+\gamma}}a \end{bmatrix},$$

and on their derivatives,

$$\begin{bmatrix} \dot{\xi}_-(0) \\ \dot{\xi}_+(0) \end{bmatrix} = \begin{bmatrix} -C_- \omega_- \sin \phi_- \\ -C_+ \omega_+ \sin \phi_+ \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and requires $\phi_- = \phi_+ = 0$. As already pointed out, for a very large middle mass, $m_1 \gg m_2$, we have $\gamma = \sqrt{m_2/m_1}$ small. The conditions on C_- and C_+ are then

$$\begin{aligned}
 C_- &\approx \frac{a\sqrt{m_1}}{\sqrt{2}}, \\
 C_+ &\approx -\frac{a\sqrt{m_1}}{\sqrt{2}}.
 \end{aligned}$$

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Putting all this together, the motion is then given by

$$\begin{aligned}
 \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} &\approx \frac{1}{\sqrt{2m_1}} \begin{bmatrix} 1 & -1 \\ \gamma^{-1} & \gamma^{-1} \end{bmatrix} \begin{bmatrix} \xi_- \\ \xi_+ \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \begin{bmatrix} \xi_- - \xi_+ \\ \gamma^{-1}(\xi_- + \xi_+) \end{bmatrix} \\
 &= \frac{1}{\sqrt{2m_1}} \frac{a\sqrt{m_1}}{\sqrt{2}} \begin{bmatrix} \cos \omega_- t + \cos \omega_+ t \\ \gamma^{-1}(\cos \omega_- t - \cos \omega_+ t) \end{bmatrix} \\
 &= a \begin{bmatrix} \cos \frac{1}{2}(\omega_+ - \omega_-)t & \cos \frac{1}{2}(\omega_+ + \omega_-)t \\ \gamma^{-1} \sin \frac{1}{2}(\omega_+ - \omega_-)t & \sin \frac{1}{2}(\omega_+ + \omega_-)t \end{bmatrix}.
 \end{aligned}$$

Indeed, we find the characteristic beat pattern as discussed in section 23 because $\omega_- \approx \omega_+$, with one of η_1 and η_2 oscillating with maximum amplitude when the other one is stationary.

4.4 This problem is Example 7.6 in Taylor. Let θ be the angle the bead makes with vertical in the hoop.

a) The kinetic energy of the particle is

$$T = \frac{m}{2} (\Omega^2 a^2 \sin^2 \theta + a^2 \dot{\theta}^2)$$

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The potential energy is

$$U = -mga \cos \theta$$

The Euler-Lagrange equation of motion is

$$\frac{d}{dt} (ma^2 \dot{\theta}) - ma^2 \Omega^2 \sin \theta \cos \theta + mga \sin \theta = 0$$

$$\ddot{\theta} = \Omega^2 \sin \theta \cos \theta - (g/a) \sin \theta$$

To get a first integral simply multiply both sides of this equation by $\dot{\theta}$. One obtains

$$\frac{d}{dt} \left(\frac{\dot{\theta}^2}{2} \right) - \frac{d}{dt} \left[\frac{\Omega^2 \sin^2 \theta}{2} + \frac{g}{a} \cos \theta \right] = 0$$

$$\frac{\dot{\theta}^2}{2} - \left[\frac{\Omega^2 \sin^2 \theta}{2} + \frac{g}{a} \cos \theta \right] = I$$

b) Possible equilibria are found by putting the RHS of the equation of motion to zero

$$\sin \theta = 0 \rightarrow \theta = 0, \pi$$

$$\cos \theta = \frac{g}{\Omega^2 a}, \Omega^2 > \frac{g}{a}$$

For $\Omega^2 < g/a$, there are two equilibria. The one at $\theta = \pi$, at the top of the hoop, is unstable. The one at $\theta = 0$, is stable. Once $\Omega^2 > g/a$, there are three equilibria. The ones at $\theta = 0, \pi$ are now both unstable (in the case $\theta = 0$, the centrifugal potential now exceeds the gravitation potential from the curved wire). The ones at $\theta_e = \pm \cos^{-1}(g/\Omega^2 a)$ are stable and arise from a balance along the wire between the centrifugal force in the rotating frame and the gravitational force in the opposite direction.

For $\Omega^2 < g/a$ and for small oscillations around $\theta = 0$

$$\frac{d^2}{dt^2} (0 + \delta\theta) = \delta\ddot{\theta} \approx \left[\Omega^2 - \frac{g}{a} \right] \delta\ddot{\theta}$$

The oscillation angular frequency is $\sqrt{g/a - \Omega^2}$.

For $\Omega^2 > g/a$ and for small oscillations around θ_e ,

$$\frac{d^2}{dt^2} (\theta_e + \delta\theta) = \delta\ddot{\theta}$$

$$= \Omega^2 ((\sin \theta_e \cos \delta\theta + \cos \theta_e \sin \delta\theta)(\cos \theta_e \cos \delta\theta - \sin \theta_e \sin \delta\theta)) - \frac{g}{a} (\sin \theta_e \cos \delta\theta + \cos \theta_e \sin \delta\theta)$$

$$\approx \Omega^2 \sin \theta_e \cos \theta_e - \frac{g}{a} \sin \theta_e + \left[\Omega^2 \cos^2 \theta_e - \frac{g}{a} \cos \theta_e \right] \delta\theta - [\Omega^2 \sin^2 \theta_e] \delta\theta$$

$$\approx 0 + 0 - [\Omega^2 \sin^2 \theta_e] \delta\theta$$

The small oscillation frequency is clearly $\Omega \sin \theta_e$, which, because $\sin^2 \theta_e = 1 - \cos^2 \theta_e$, is also $\sqrt{\Omega^2 - g/a}$.

c) The conjugate momentum is

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = ma^2 \dot{\theta}$$

The Hamiltonian is

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$$\begin{aligned}
 H &= p_\theta \frac{p_\theta}{ma^2} - L = \frac{p_\theta^2}{2ma^2} - \frac{m}{2} \Omega^2 a^2 \sin^2 \theta - mga \cos \theta \\
 &= \frac{p_\theta^2}{2ma^2} + ma^2 \left(I - \frac{\dot{\theta}^2}{2} \right) = ma^2 I,
 \end{aligned}$$

which is a constant of the motion. However, it is not the total energy

$$T + V = \frac{p_\theta^2}{2ma^2} + \frac{m}{2} \Omega^2 a^2 \sin^2 \theta - mga \cos \theta = m\Omega^2 a^2 \sin^2 \theta + ma^2 I,$$

which increases from the rotation as the particle moves away from the axis of rotation.

a) The Lagrangian in polar coordinates in the orbital plane is

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - V(r).$$

The Euler-Lagrange equation in r is

$$\begin{aligned}\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} &= \frac{\partial L}{\partial r} \\ m\ddot{r} &= mr\dot{\phi}^2 - V'(r).\end{aligned}$$

At the equilibrium, $r = r_0$ and $\phi = \Omega t$, this gives $V'(r_0) = mr_0\Omega^2$.

With $r = r_0 + \eta_r$ and $\phi = \Omega t + \eta_\phi$, we can expand the Lagrangian:

$$\begin{aligned}L &= \frac{1}{2}m(\dot{\eta}_r^2 + (r_0 + \eta_r)^2(\Omega + \dot{\eta}_\phi)^2) - V(r_0 + \eta_r) \\ &= \frac{1}{2}m(\dot{\eta}_r^2 + (r_0^2 + 2r_0\eta_r + \eta_r^2)(\Omega^2 + 2\Omega\dot{\eta}_\phi + \dot{\eta}_\phi^2)) - V(r_0 + \eta_r) \\ &= \frac{1}{2}m(\dot{\eta}_r^2 + r_0^2\Omega^2 + 2r_0\Omega^2\eta_r + \Omega^2\eta_r^2 + 2r_0^2\Omega\dot{\eta}_\phi + 4r_0\Omega\eta_r\dot{\eta}_\phi + 2\Omega\dot{\eta}_\phi\eta_r^2 + r_0^2\dot{\eta}_\phi^2 + 2r_0\eta_r\dot{\eta}_\phi^2 + \eta_r^2\dot{\eta}_\phi^2) \\ &\quad - V(r_0) - \eta_r V'(r_0) - \frac{1}{2}\eta_r^2 V''(r_0) + \mathcal{O}(\eta^3) \\ &= \frac{1}{2}m(\dot{\eta}_r^2 + r_0^2\Omega^2 + \Omega^2\eta_r^2 + 2r_0^2\Omega\dot{\eta}_\phi + 4r_0\Omega\eta_r\dot{\eta}_\phi + 2\Omega\dot{\eta}_\phi\eta_r^2 + r_0^2\dot{\eta}_\phi^2 + 2r_0\eta_r\dot{\eta}_\phi^2 + \eta_r^2\dot{\eta}_\phi^2) \\ &\quad - V(r_0) - \frac{1}{2}\eta_r^2 V''(r_0) + \mathcal{O}(\eta^3),\end{aligned}$$

where we used the previously found expression for $V'(r_0)$.

Since the Lagrangian is cyclic in ϕ , the angular momentum $\ell = p_\phi$ is conserved. After expansion,

$$\begin{aligned}\ell = p_\phi = \frac{\partial L}{\partial \dot{\phi}} &= mr^2\dot{\phi} \\ &= m(r_0 + \eta_r)^2(\Omega + \dot{\eta}_\phi) \\ &= mr_0^2\Omega + mr_0^2\dot{\eta}_\phi + 2mr_0\Omega\eta_r + 2mr_0\eta_r\dot{\eta}_\phi + m\Omega\eta_r^2 + m\eta_r^2\dot{\eta}_\phi\end{aligned}$$

the angular momentum ℓ is constant and equal to $mr_0^2\Omega$. This results in

$$r_0^2\dot{\eta}_\phi + 2r_0\eta_r\dot{\eta}_\phi + \eta_r^2\dot{\eta}_\phi = -2r_0\Omega\eta_r - \Omega\eta_r^2.$$

We can use this relation (multiplied with 2Ω) to remove the non-bilinear terms in the expanded Lagrangian, after removing constant terms:

$$\begin{aligned} L &= \frac{1}{2}m(\dot{\eta}_r^2 + \Omega^2\eta_r^2 + 2r_0^2\Omega\dot{\eta}_\phi + 4r_0\Omega\eta_r\dot{\eta}_\phi + 2\Omega\dot{\eta}_\phi\eta_r^2 + r_0^2\dot{\eta}_\phi^2 + 2r_0\eta_r\dot{\eta}_\phi^2 + \eta_r^2\dot{\eta}_\phi^2) - \frac{1}{2}\eta_r^2V''(r_0) + \mathcal{O}(\eta^3) \\ &= \frac{1}{2}m(\dot{\eta}_r^2 + r_0^2\dot{\eta}_\phi^2 - 3\Omega^2\eta_r^2) - \frac{1}{2}\eta_r^2V''(r_0) + \mathcal{O}(\eta^3). \end{aligned}$$

This is in the form $L = \frac{1}{2}\dot{\eta}^T M \dot{\eta} - \frac{1}{2}\eta^T V \eta$ with

$$\begin{aligned} M &= m \begin{bmatrix} 1 & 0 \\ 0 & r_0^2 \end{bmatrix}, \\ V &= \begin{bmatrix} V''(r_0) + 3m\Omega^2 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

The eigenvalue equation $\det(V - \omega^2 M) = 0$ becomes then

$$\begin{aligned} \begin{vmatrix} V''(r_0) + 3m\Omega^2 - m\omega^2 & 0 \\ 0 & -mr_0^2\omega^2 \end{vmatrix} &= 0, \\ -mr_0^2\omega^2 (V''(r_0) + 3m\Omega^2 - m\omega^2) &= 0. \end{aligned}$$

The non-zero eigenvalue of this system is

$$\omega^2 = 3\Omega^2 + \frac{1}{m}V''(r_0) = \Omega^2 \left(3 + r_0 \frac{V''(r_0)}{V'(r_0)} \right).$$

This eigenvalue is positive, and the equilibrium therefore stable, as long as the following criterion is satisfied:

$$V''(r_0) + \frac{3}{r_0}V'(r_0) > 0.$$

b) We can derive the same criterium from the one-dimensional Lagrangian with an effective potential $V_{eff}(r)$

$$V_{eff}(r) = V(r) + \frac{\ell^2}{2mr^2}.$$

Note the difference with (3.9) where the energy E is considered. The criterion for stability at the equilibrium is now, with $\ell = mr^2\Omega$ as found above,

$$\begin{aligned} V_{eff}''(r_0) &> 0 \\ V''(r_0) + \frac{3\ell^2}{mr_0^4} &> 0 \\ V''(r_0) + 3m\Omega^2 &> 0 \\ V''(r_0) + \frac{3}{r_0}V'(r_0) &> 0, \end{aligned}$$

4.6

where we used again the expression for $V'(r_0)$.

c) If $V(r) = -\lambda r^{-n}$, then the criterion becomes

$$-n(n+1)\lambda r_0^{-n-2} + 3n\lambda r_0^{-n-2} > 0.$$

This is satisfied for $n(n+1) - 3n < 0$ or $n+1 < 3$ or $n < 2$.

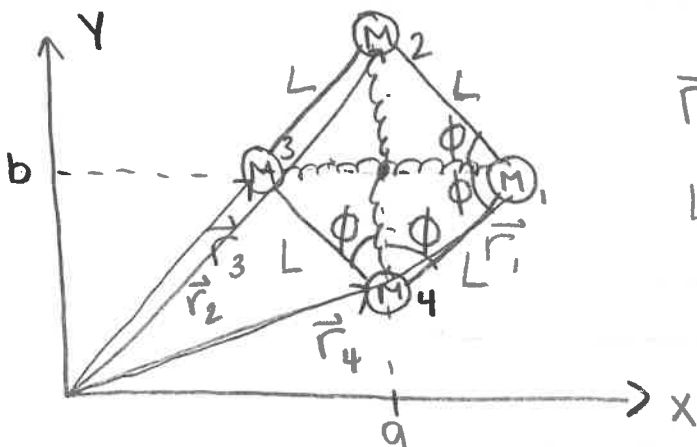
d) If $V(r) = -\frac{\lambda}{r} \exp(-\frac{r}{a})$, then the criterion becomes

$$\begin{aligned} & \left(-\frac{1}{a^2} \frac{\lambda}{r_0} \exp(-\frac{r_0}{a}) - \frac{1}{a} \frac{\lambda}{r_0^2} \exp(-\frac{r_0}{a}) - 2 \frac{\lambda}{r_0^3} \exp(-\frac{r_0}{a}) - \frac{1}{a} \frac{\lambda}{r_0^2} \exp(-\frac{r_0}{a}) \right) \\ & \quad + 3 \frac{1}{r_0} \left(\frac{1}{a} \frac{\lambda}{r_0} \exp(-\frac{r_0}{a}) + \frac{\lambda}{r_0^2} \exp(-\frac{r_0}{a}) \right) > 0 \\ & \left(-\frac{1}{a^2} - \frac{1}{ar_0} - 2 \frac{1}{r_0^2} - \frac{1}{ar_0} + 3 \frac{1}{ar_0} + 3 \frac{1}{r_0^2} \right) \frac{\lambda}{r_0} \exp(-\frac{r_0}{a}) > 0 \\ & \left(\frac{1}{r_0^2} + \frac{1}{ar_0} - \frac{1}{a^2} \right) > 0 \\ & 1 + \frac{r_0}{a} - \left(\frac{r_0}{a} \right)^2 > 0 \\ & \left(\frac{\sqrt{5}-1}{2} + \frac{r_0}{a} \right) \left(\frac{\sqrt{5}+1}{2} - \frac{r_0}{a} \right) > 0. \end{aligned}$$

This is satisfied if r_0/a is smaller than the golden ratio, or

$$r_0 < \frac{\sqrt{5}+1}{2} a.$$

Fetter and Walecka 4.8:



$$\vec{r}_{cm} = \vec{a}x + \vec{b}y$$

L is fixed

$$\begin{aligned} \vec{r}_1 &= (a + L \cos \phi) \hat{x} + b \hat{y} \\ \vec{r}_2 &= a \hat{x} + (b + L \sin \phi) \hat{y} \\ \vec{r}_3 &= (a - L \cos \phi) \hat{x} + b \hat{y} \\ \vec{r}_4 &= a \hat{x} + (b - L \sin \phi) \hat{y} \end{aligned}$$

Kinetic Energy:

$$\begin{aligned} \dot{\vec{r}}_1 &= -L \dot{\phi} \sin \phi \hat{x} + 0 \hat{y} \\ \dot{\vec{r}}_2 &= 0 \hat{x} + L \dot{\phi} \cos \phi \hat{y} \\ \dot{\vec{r}}_3 &= L \dot{\phi} \sin \phi \hat{x} + 0 \hat{y} \\ \dot{\vec{r}}_4 &= 0 \hat{x} - L \dot{\phi} \cos \phi \hat{y} \end{aligned}$$

$$\begin{aligned} |\dot{\vec{r}}_1|^2 &= L^2 \dot{\phi}^2 \sin^2 \phi \\ |\dot{\vec{r}}_2|^2 &= L^2 \dot{\phi}^2 \cos^2 \phi \\ |\dot{\vec{r}}_3|^2 &= L^2 \dot{\phi}^2 \sin^2 \phi \\ |\dot{\vec{r}}_4|^2 &= L^2 \dot{\phi}^2 \cos^2 \phi \end{aligned}$$

$$T = \frac{1}{2} \sum_{i=1}^4 M_i \dot{\vec{r}}_i^2 = \frac{1}{2} M \sum_{i=1}^4 |\dot{\vec{r}}_i|^2 = \frac{1}{2} M L^2 \dot{\phi}^2 (2(\sin^2 \phi + \cos^2 \phi))$$

$$T = \frac{1}{2} M L^2 \dot{\phi}^2 = M L^2 \dot{\phi}^2$$

Potential Energy:

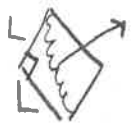
$$U = \text{Spring potentials} = \frac{1}{2} K (\text{spring stretch})^2$$

$$\begin{aligned} \text{spring length}_{24} &= |\vec{r}_2 - \vec{r}_4| = |b + L \sin \phi - b + L \sin \phi| \\ \text{spring length}_{24} &= 2L \sin \phi \end{aligned}$$

$$\begin{aligned} \text{Spring length}_{13} &= |\vec{r}_1 - \vec{r}_3| = |a + L \cos \phi - a + L \cos \phi| \\ \text{spring length}_{13} &= 2L \cos \phi \end{aligned}$$

continued $\rightarrow$

Spring stretch/compress = Spring length - length initial



length initial is when configuration is in square configuration

$$\therefore \text{length initial} = \sqrt{2}L \quad \{ = \sqrt{L^2 + L^2} \}$$

$$\text{stretch/compress } 13 = 2L \cos \phi - \sqrt{2}L$$

$$\text{stretch/compress } 24 = 2L \sin \phi - \sqrt{2}L$$

$$U_{13} = \frac{1}{2} k (2L \cos \phi - \sqrt{2}L)^2$$

$$U_{24} = \frac{1}{2} k (2L \sin \phi - \sqrt{2}L)^2$$

$$U_{13} = \frac{1}{2} k L^2 (2 \cos \phi - \sqrt{2})^2$$

$$= \frac{1}{2} k L^2 (4 \cos^2 \phi - 2\sqrt{2} \cdot 2 \cos \phi + 2)$$

$$= \frac{1}{2} k L^2 \cdot 2 (2 \cos^2 \phi - 2\sqrt{2} \cos \phi + 1)$$

$$= k L^2 (2 \cos^2 \phi - 2\sqrt{2} \cos \phi + 1)$$

$$U_{24} = \frac{1}{2} k L^2 (2 \sin \phi - \sqrt{2})^2$$

$$= \frac{1}{2} k L^2 (4 \sin^2 \phi - 2\sqrt{2} \cdot 2 \sin \phi + 2)$$

$$= k L^2 (2 \sin^2 \phi - 2\sqrt{2} \sin \phi + 1)$$

$$U = U_{13} + U_{24} = k L^2 (2(\sin^2 \phi + \cos^2 \phi) + 2 - 2\sqrt{2}(\sin \phi + \cos \phi))$$

$$= k L^2 (4 - 2\sqrt{2}(\sin \phi + \cos \phi))$$

$$U = 2k L^2 (2 - \sqrt{2}(\sin \phi + \cos \phi))$$

$$\mathcal{L} = T - U$$

$$\boxed{\mathcal{L} = M L^2 \dot{\phi}^2 - 2k L^2 (2 - \sqrt{2}(\sin \phi + \cos \phi))}$$

Equation of Motion:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \phi}$$

continued $\rightarrow$

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2ML^2\dot{\phi} \rightarrow \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2ML^2\ddot{\phi}$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = 2KL^2\sqrt{2}(\cos\phi - \sin\phi)$$

$$2ML^2\ddot{\phi} = 2KL^2\sqrt{2}(\cos\phi - \sin\phi)$$

$$\boxed{M\ddot{\phi} = K\sqrt{2}(\cos\phi - \sin\phi)}$$

In small oscillations: assume only small change from equilibrium
 * in equilibrium $\phi = 45^\circ$ since it is a square configuration at equilibrium.

$$\phi' = 45^\circ + \delta \text{ \& small change}$$

$$\ddot{\phi}' = \ddot{\delta}$$

$$M\ddot{\phi}' = K\sqrt{2}(\cos(\phi') - \sin(\phi'))$$

$$M\ddot{\delta} = K\sqrt{2}(\cos(45^\circ + \delta) - \sin(45^\circ + \delta))$$

Trig identities: $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
 $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$$M\ddot{\delta} = K\sqrt{2}(\cos 45^\circ \cos \delta - \sin 45^\circ \sin \delta - \sin 45^\circ \cos \delta - \cos 45^\circ \sin \delta)$$

$$\cos(45^\circ) = \sin(45^\circ) = \frac{\sqrt{2}}{2}$$

$$M\ddot{\delta} = K\sqrt{2}\left(\frac{\sqrt{2}}{2}\cos\delta - \frac{\sqrt{2}}{2}\sin\delta - \frac{\sqrt{2}}{2}\cos\delta - \frac{\sqrt{2}}{2}\sin\delta\right)$$

$$M\ddot{\delta} = -K\sqrt{2} \cdot \sqrt{2}\left(\frac{2}{2}\sin\delta\right)$$

$$M\ddot{\delta} = -2k \sin \delta$$

$$\ddot{\delta} = -\frac{2k}{M} \sin \delta$$

$$\ddot{\delta} = -\omega_0^2 \sin \delta = -\omega_0^2 \delta \quad \swarrow$$

small δ approximation

$$\omega_0^2 = \frac{2k}{M}$$

$$\boxed{\omega_0 = \sqrt{\frac{2k}{M}}}$$